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Bihar Engineering University, Patna

Examination 2013

Time: 3 hours

Code: 105314 ***Subject: Operating Systems*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are TEN questions in this paper.
(iii) Attempt any FIVE questions.

Q.1 Solve all questions : **[14]**

(a) Define operating system. What are the duties performed by an operating system?

(b) What are the main advantages of time sharing systems?

(c) What are system calls? Explain their importance.

Q.2 Solve both questions : **[14]**

(a) Differentiate between a file and a Directory. Justify for the need of file system.

(b) Explain the main features of general graph directory structure.

Q.3 Solve both questions : **[14]**

(a) What do you understand by deadlock? Discuss the methods to avoid deadlock in an operating system.

(b) Explain the working of resource allocation graph.

Q.4 Solve both questions : **[14]**

(a) What is the difference between physical and logical address space?

(b) What are semaphores? How can they be used to handle the process synchronization problem?

Q.5 Solve both questions : **[14]**

(a) What are the benefits of segmentation?

(b) Explain the segmentation technique with the help of a diagram.

Q.6 Solve both questions : **[14]**

(a) Explain the following terms:
(i) Turnaround time
(ii) Waiting time
(iii) Response time
(iv) Throughput

(b) Consider the following page replacement string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1
Find the number of page faults using the following algorithms for a memory of three frames:
(i) FIFO algorithm
(ii) LRU algorithm

Q.7 Solve both questions : **[14]**

(a) What do you understand by fragmentation? Explain the difference between external and internal fragmentations.

(b) What is a thread? Explain the benefits of using threads.

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Q.8 Solve both questions : **[14]**

(a) What do you understand by address binding? Explain the various address binding schemes.

(b) Explain the importance of virtual memory.

Q.9 Solve both questions : **[14]**

(a) What do you understand by disk scheduling? Describe seek time and rotational latency.

(b) Describe SCAN scheduling. How is C-SCAN different from SCAN?

Q.10 Write short notes on (any three): **[14]**

(a) Micro kernel

(b) Bankers Algorithm

(c) Process control block

(d) Paging

(e) Indexed allocation

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